

UENO, Japan

WMO: 476490

Lat: 34.7617N

Lon: 136.1419E

Elev: 161

StdP: 99.41

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	-2.1	-9.1	1.8	3.9	-7.5	2.0	3.6	9.9	2.8	9.1	3.2	1.6	160	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	34.2	24.7	33.0	24.5	31.7	24.1	25.9	31.2	25.4	30.4	24.9	29.7	3.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.1	27.4	24.2	19.4	27.3	23.6	18.8	27.1	81.0	31.2	79.0	30.5	77.0	29.8	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.1	6.2	DB	-5.1	35.8	1.1	0.9	-5.9	36.5	-6.5	37.0	-7.2	37.5	-7.9	38.1	
			WB	-5.8	26.7	1.1	0.7	-6.6	27.1	-7.2	27.5	-7.9	27.9	-8.7	28.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	3.6	4.3	7.5	12.9	18.2	22.1	26.4	27.4	23.4	17.2	11.0	5.9
	DBStd	8.70	2.02	2.57	2.98	3.32	2.53	2.40	2.23	1.86	2.89	2.96	3.09	2.57
	HDD10.0	616	198	162	91	11	0	0	0	0	0	24	130	
	HDD18.3	2048	456	394	337	164	33	1	0	2	57	220	385	
	CDD10.0	2461	0	1	12	98	254	362	509	541	401	224	54	3
	CDD18.3	852	0	0	0	2	29	113	251	282	152	22	0	0
	CDH23.3	7166	0	0	0	32	243	691	2274	2813	1023	90	0	0
	CDH26.7	2636	0	0	0	3	44	179	914	1187	301	8	0	0
Wind	WSAvg	2.5	3.0	2.9	2.9	2.7	2.5	2.3	2.2	2.3	2.3	2.1	2.2	2.7
Precipitation	PrecAvg	1498	60	71	106	109	151	196	188	133	200	127	86	60
	PrecMax	1840	141	127	187	211	360	345	383	375	416	306	232	115
	PrecMin	1019	17	24	49	34	54	75	41	33	37	20	10	7
	PrecStd	232	30	28	37	44	75	67	92	80	98	62	53	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.6	20.9	26.5	30.1	32.7	35.2	35.9	33.2	27.7	21.7	16.5
		MCWB	8.7	11.1	12.0	15.7	18.0	23.4	24.9	25.0	24.3	21.2	15.8	12.7
	2%	DB	11.0	13.5	18.2	23.8	27.8	30.4	34.0	34.5	31.4	25.6	19.9	14.2
		MCWB	6.9	8.5	10.8	14.2	17.4	21.8	24.7	24.8	23.5	19.4	14.6	10.2
	5%	DB	9.5	11.4	15.8	21.8	26.0	28.7	32.8	33.5	29.9	24.0	18.3	12.6
		MCWB	5.8	7.1	10.2	13.7	16.8	20.7	24.5	24.7	22.9	18.3	14.0	9.1
	10%	DB	8.1	9.6	13.9	19.8	24.3	27.2	31.4	32.2	28.4	22.6	16.7	11.2
		MCWB	4.9	6.2	9.4	13.2	16.4	20.4	24.2	24.4	22.2	17.6	13.3	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.6	14.7	19.2	21.7	25.0	26.3	26.7	25.7	23.0	17.7	13.7
		MCDB	11.9	15.6	16.9	21.8	25.1	29.4	32.1	32.0	29.9	25.7	19.5	15.2
	2%	WB	7.9	10.0	13.0	17.0	20.0	23.8	25.7	26.1	24.7	21.2	16.2	11.4
		MCDB	9.7	11.8	16.1	20.5	23.4	27.8	31.2	31.3	28.8	24.0	18.1	13.3
	5%	WB	6.4	8.1	11.4	15.6	18.9	22.9	25.2	25.6	24.1	19.8	15.1	9.7
		MCDB	8.9	10.2	14.4	19.4	22.9	26.6	30.4	30.7	28.0	22.5	17.3	11.7
	10%	WB	5.1	6.5	9.9	14.3	17.9	21.9	24.7	25.2	23.4	18.6	13.9	8.2
		MCDB	7.5	9.2	13.1	18.0	21.9	25.3	29.5	30.0	26.9	21.4	16.3	10.7
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.0	10.6	11.6	10.8	8.6	8.5	9.1	8.5	9.4	9.9	8.9
		MCDBR	11.1	12.9	14.6	15.3	14.7	11.6	10.7	11.1	10.1	10.9	11.6	11.4
		MCWBR	7.4	8.3	8.6	7.3	5.9	4.3	3.4	3.3	3.8	5.3	7.1	7.7
	5% WB	MCDBR	9.9	10.5	11.4	10.8	10.2	8.6	9.3	9.5	8.3	8.2	9.1	9.4
		MCWBR	7.4	7.9	8.1	6.6	5.5	3.8	3.4	3.2	3.5	4.6	6.4	7.3
Clear-Sky Solar Irradiance	taub	0.399	0.438	0.497	0.534	0.540	0.559	0.559	0.527	0.487	0.450	0.413	0.394	
	taud	2.145	2.052	1.915	1.875	1.925	1.950	2.002	2.081	2.162	2.183	2.192	2.174	
	Ebn at Noon	775	787	772	767	767	750	747	764	776	769	755	753	
	Edh at Noon	119	146	182	199	191	186	176	160	141	127	113	110	
All-Sky Solar Radiation	RadAvg	2.33	2.99	3.96	4.77	5.17	4.64	4.93	5.06	3.99	3.21	2.52	2.11	
	RadStd	0.16	0.24	0.28	0.45	0.56	0.44	0.66	0.45	0.35	0.33	0.22	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page